Dr. Georgii Oblapenko

Born on 10 Feb. 1992

Chair of Applied & Computational Mathematics (ACoM), RWTH Aachen, Schinkelstraße 2, 52062 Aachen, Germany

- +49-174-476-9489
- oblapenko@acom.rwth-aachen.de
- LinkedIn: George-Oblapenko
- ResearchGate: Georgii-Oblapenko
- ORCID: 0000-0002-1434-7166
- Google Scholar: Georgii Oblapenko
- Github: knstmrd
- Languages: English (fluent), German (fluent), Russian (native speaker)

Researcher with 10+ years of experience in the field of numerical model development for simulation of non-equilibrium flows.

Research interests

- Mathematical modelling of multi-species non-equilibrium flows
- Particle-based methods for rarefied flows
- Discrete velocity methods
- Entropy-stable high-order discretizations
- Optimization in context of kinetic theory
- Research software development

Employment history

Oct. 2026 – April 2027 **Interim Professor for Industrial Mathematics (W2)** • RWTH Aachen, Chair of Applied and Computational Mathematics.
Teaching of "Fundamentals of Mathematics I" for Bachelor students of the "Computational Engineering and Sciences" programme. Further development of kinetic methods for modelling non-equilibrium flows.

Nov. 2023 – **Present** **Member of Scientific Staff** • RWTH Aachen, Chair of Applied and Computational Mathematics.
Development of 1) kinetic methods for modelling of non-equilibrium flows in continuum and rarefied regimes 2) entropy-stable DG schemes for multi-component real gas flows with thermo-chemical non-equilibrium. Development and teaching of new courses for Master students, teaching of global exercises for Bachelor students.

Jan. 2023 – Sep. 2023 **Scientific Consultant** • UT Austin, Oden Institute for Computational Engineering and Sciences / Dept. of Aerospace Engineering and Engineering Mechanics.
Development of new PIC-DSMC methods. Co-supervision of Master student developing PIC-DSMC methods for high-voltage gap closure modeling.

Employment history (continued)

Nov. 2021 – Oct. 2023 **Postdoctoral Researcher** • German Aerospace Center (DLR), Institute of Aerodynamics and Flow Technology, Spacecraft Department.
Uncertainty quantification of hypersonic expanding continuum flows. Development of 1) an efficient photon Monte Carlo radiation transport code 2) novel kinetic methods for simulations of plasma and gases with internal degrees of freedom 3) low-rank methods for the Boltzmann equation. Supported by a personal grant of the Alexander von Humboldt Foundation.

July 2018 – June 2021 **Postdoctoral Researcher** • UT Austin, Oden Institute for Computational Engineering and Sciences / Dept. of Aerospace Engineering and Engineering Mechanics.
Development of 1) hybrid numerical methods for kinetic problems 2) new algorithms for rarefied plasma simulations 3) novel kinetic models for gases with internal degrees of freedom and chemically reacting gas flows.

Sept. 2013 – Dec. 2017 **Research engineer** • Saint-Petersburg State University, Department of Hydroaeromechanics.
Lead developer of a C++ library for kinetic theory computations and coupling with CFD codes. Development of 1) theoretical models of reaction rates in viscous gas flows and their numerical modeling 2) simplified models for vibrational relaxation rates.

Education

2015 – 2017 **Ph.D., Saint-Petersburg State University** in Physical and Mathematical Sciences.
Area of specialization: Mechanics of Fluids, Gases and Plasma.
Thesis title: *Physico-chemical relaxation rates in viscous non-equilibrium gas flows*.
Supervisor: Prof. Kustova E.V.

2013 – 2015 **M.Sc. (summa cum laude), Saint-Petersburg State University** in Mechanics and Mathematical Modeling.
Area of specialization: Molecular-kinetic theory of fluids and gases.
Thesis title: *Vibrational relaxation rates in viscous rarefied gas flows*.
Supervisor: Prof. Kustova E.V.

2009 – 2013 **B.Sc., Saint-Petersburg State University** in Mathematics and Mechanics.
Supervisor: Prof. Kustova E.V.

Grants and awards

Personal funding and travel grants

2026 – 2029 **DFG individual grant**. “Simulation of rare and transient events in rarefied gas dynamics”. Funding of a doctoral position for 3 years (ca. 330000 EUR)

2021 – 2023 **Alexander von Humboldt Foundation** Research Stipend for Postdoctoral Researchers (60000 EUR).

2017 – 2018 **German Academic Exchange Service (DAAD)** Research Stipend (3300 EUR).

2016 **Santander Bank** Research Stipend (1400 EUR).

2015 – 2016 **Russian Foundation for Basic Research** Research Stipend (4500 EUR).

2013, 2014, 2017 **Saint-Petersburg State University** Travel funding awards.

Awards

2023 **Marie Skłodowska-Curie Postdoctoral Action Seal of Excellence**

Grants and awards (continued)

2017 **Stipend** of the Russian President for students and PhD students in prioritized study programmes (3800 EUR).

2016 **Stipend** of the Russian Government for students and PhD students in prioritized study programmes (1900 EUR).

Stipend of the Russian Government for students and PhD students (500 EUR).

2013 **Winner** of the Saint-Petersburg Government grant competition for undergraduate and graduate students (500 EUR).

Publications

The full list of publications consists of **30** (including accepted papers) peer-reviewed publications in SCOPUS/Web of Science-indexed journals and conference proceedings, as well as **22** publications without peer review (conference proceedings, preprints).

Publication statistics (as of August 2026):

	SCOPUS	**Google Scholar**
Number of publications	36	67
h-index	11	13
Citations	404	669

Conference and seminar presentations

Conferences — Presented at **26** international and **11** national (German, Russian) conferences.

Plenary talks — **2** invited plenary lectures at international conferences and workshops.

Invited seminar talks — **27** invited talks at universities and research institutes in the USA, Germany, Belgium, France, and the Netherlands.

Public outreach and pop-science talks — Lightning Math Night (RWTH Aachen, 2026) ○ Exhibition "Falling Rhythm" (Barcelona, Spain & Online, 2025)

Teaching and student supervision

Teaching: lectures

Winter semester 2026/2027 — Teaching of the course "Foundations of Mathematics I", Bachelor Study Program "Computational Engineering Science", RWTH Aachen.

Summer semester 2026 — Development and teaching of new Master course on "Modern Simulation Software Development", RWTH Aachen (together with Dr. Lambert Theisen). Course offered to Master and Bachelor students of the Computational Engineering Science and Simulation Sciences programs.

Winter semester 2025/2026 — Invited lecture on kinetic theory in the framework of the Master course on "Aerospace Fundamentals", Justus Liebig University Gießen.

Summer semester 2025 — Development and teaching of new Master course on "Particle-based Simulation Methods", RWTH Aachen. Course offered to Master and Bachelor students of the Mathematics, Computational Engineering Science, and Simulation Sciences programs.

Winter semester 2024/2025 — Invited lecture on kinetic theory in the framework of the Master course on "Aerospace Fundamentals", Justus Liebig University Gießen.

Teaching and student supervision (continued)

Winter semester 2023/2024	Invited lecture on kinetic theory in the framework of the Master course on "Aerospace Fundamentals", Justus Liebig University Gießen.

Teaching: exercises

Winter semester 2025/2026	Global Exercise for "Foundations of Mathematics I", Bachelor Study Program "Computational Engineering Science", RWTH Aachen.
Summer semester 2024	Global Exercise for "Foundations of Mathematics II", Bachelor Study Program "Computational Engineering Science", RWTH Aachen.
Winter semester 2023/2024	Global Exercise for "Foundations of Mathematics I", Bachelor Study Program "Computational Engineering Science", RWTH Aachen.

Additional teaching qualifications

2026	Enrolled in program for "Extensions Module" certificate of the program "Professionelle Lehrkompetenz für die Hochschule", Hochschuldidaktik NRW, expected ending date: August 2026
2025	"Basics Module" certificate of the program "Professionelle Lehrkompetenz für die Hochschule", Hochschuldidaktik NRW
2024–2026	15 Completed courses at the Center for Teaching and Learning Services, RWTH Aachen.
2020	Completed Title IX Training on prevention and reporting of sexual discrimination and harassment, UT Austin.

Supervision, thesis reviews

PhD supervision	**2026 – Present** Supervision of PhD Student Woohyuk Noh. **2024 – Present** Co-supervision of PhD Student Eda Yilmaz, joint publication in SIAM J. Appl. Math. and in peer-reviewed conference proceedings.
Master student supervision	**2026 – Present** Supervision of a Master student at RWTH Aachen, conference paper in preparation. **2022 – 2023** Co-supervision of a Master student at UT Austin, resulting in a peer-reviewed conference proceedings publication.
Bachelor student supervision	**2025 – Present 2** Bachelor theses at RWTH Aachen, resulting in one peer-reviewed conference proceedings publication; second conference paper in preparation.
Research assistants	**2025 – Present 4** student research assistants at RWTH Aachen
Research projects, seminars	**2025** Supervision of a group research project at RWTH Aachen.
Referee of theses	**2024 – Present** Referee of **4** Bachelor and **1** Master theses, RWTH Aachen. **2020 – 2021** Referee of **1** Bachelor-, **2** Diploma- and **3** Master theses at Saint-Petersburg State University and Peter the Great St. Petersburg Polytechnic University.

Professional membership and other activities

2025 – **Present**	**RWTH hiring committee** • Member of the hiring committee for the W2 professorship “Industrial mathematics”, RWTH Aachen.
March 2022 – March 2024	**COST Network** • Member of the networking group “Mathematical models for interacting dynamics on networks” (MAT-DYN-NET), supported by the COST (European Cooperation in Science and Technology) program.
July 2021 – **Present**	**RGD NextGen Group** • Member of the RGD NextGen Group, aimed at increasing outreach of the Rarefied Gas Dynamics community.
Spring 2021 – Jan. 2024	**American Institute of Aeronautics and Astronautics (AIAA) Thermophysics Technical Committee** • Member of Education, Emerging Technologies, Conference subcommittees.
Dec. 2019 – Jan. 2024	**American Institute of Aeronautics and Astronautics (AIAA)** • Senior AIAA Member .

Editing and review activities

Editor	Proceedings of the 33rd International Symposium on Rarefied Gas Dynamics, Springer, 2026 (co-editor), `ISBN 978-3-032-00093-1`.
Funding Proposal reviewer	DFG (since 2026).
Journal reviewer	Computers and Fluids (Elsevier) · Acta Astronautica (Elsevier) · Journal of Computational Physics (Elsevier) · Computer Methods in Applied Mechanics and Engineering (Elsevier) · International Journal of Heat and Mass Transfer (Elsevier) · Applied Mathematical Modelling (Elsevier) · Physics of Fluids (AIP) · Physics of Plasmas (AIP) · Journal of Thermophysics and Heat Transfer (AIAA) · Journal on Scientific Computing (SIAM) · Microfluidics and Nanofluidics (Springer).
Proceedings and conference reviewer	International Symposium on Rarefied Gas Dynamics · STAB/DGLR Symposium · AIAA Scitech · AIAA Aviation

Event organisation

RGD33 Symposium	Member of the Local Advisory Committee of the “33rd International Symposium on Rarefied Gas Dynamics”, 15-19 July 2024, Göttingen, Germany.
VIII Polyakhov Readings	Member of the Local Organizing Committee of the international conference on mechanics “VIII Polyakhov Readings”, 30 January-2 February 2018, Saint-Petersburg, Russia.

Research software

Developed open-source simulation software includes

- A variable-weight Direct Simulation Monte Carlo code
- Sparse entropic quadrature method for gas-kinetic problems
- Extensions of Trixi.jl to aerodynamic applications
- Reproducible examples for the course *Modern Simulation Software Development*, RWTH Aachen
- Extensions of Trixi.jl to entropy-conservative simulations of reacting multi-species gas flows
- A library for calculation of thermodynamic and transport properties of non-equilibrium gas flows

- A tool for calculation of vibrational relaxation times based on a kinetic theory definition